MILITARY SPECIFICATION

THIONYL CHLORIDE ELECTROLYTE
(For use in Ammunition)

This specification approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions for the manufacturer of Thionyl Chloride Electrolyte Solution for use in batteries.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-L-63470	- Lithium Chloride (For Use in Ammunition)
MIL-A-63471	- Aluminum Chloride (Anhydrous)
MIL-T-63466	- Thionyl Chloride (For Use in Ammunition)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-1218	- ACS Chemicals

FSC 6810

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM-E300 Sampling Industrial Chemicals
ASTM-D2905 Statements on Number of Specimens Required to Determine the Average Quality of a Textile Material.

(Appplication for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19013).

3. REQUIREMENTS

3.1 Thionyl Chloride Electrolyte Solution. Materials used in the mixing of a Thionyl Chloride Electrolyte solution shall conform with the specifications given below. The nominal composition of the Electrolyte solution shall be Thionyl Chloride 89.5% and Lithium Aluminum Chloride (L_1, Al, Cl_4) 10.5% (See 6.3 and 6.4).

MATERIALS	APPLICABLE SPECIFICATION
Lithium Chloride	MIL-L-63470
Aluminum Chloride	MIL-A-63471
Thionyl Chloride	MIL-T-63466

3.2 Properties. The electrolyte shall conform to the following properties when tested in accordance with applicable paragraphs:

TABLE I

Requirements for Electrolyte

Properties	Requirements	Test Paragraph
Water, ppm, maximum	850	4.6.1
Iron, ppm, maximum	50	4.6.2
Density, g/mL at 24°C (± 0.01)	1.68	4.6.3

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3.3 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of Thionyl Chloride Electrolyte Solution obtained by sampling as described in 4.5.1 (smaller weight samples may be taken that are consistent with the requirements of the test). The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained in the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations for tests specified in 4.6 of this specification.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of Thionyl Chloride Electrolyte produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of

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operating conditions. Each lot shall consist of that quantity of Thionyl Chloride Electrolyte that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous (see MIL-A-48078). The product shall be submitted for inspection in accordance with MIL-STD-105.

4.5 Testing (see 6.4). PRECAUTION: This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicale safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Two (2) individual 100 ml samples from each batch or lot of the Thionyl Chloride Electrolyte solution shall be sampled as specified in ASTM E300. The sample shall be tested independently in accordance with the applicable methods given in 4.6. If either sample fails to meet any of the test requirements, the batch or lot shall be rejected. The samples shall be subjected to the tests Defect Classification listed below:

Inspection/Requirement	Defect Classification
Thionyl Chloride Electrolyte Solution	Major
Properties	Major

4.5.2 Inspection equipment. The contractor shall use commercial inspection equipment for all tests and examinations specified in para 4.6. He is also charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use.

4.6 Test methods and procedures. (See 6.2 and 6.4). The tests to determine compliance with the requirements given in Table I shall be performed on the Thionyl Chloride Electrolyte samples obtained in 4.5.1. All tests given in this section shall be performed using prescribed procedures for replicate determinations given in standard analytical chemistry textbooks or ASTM-D2905. Also, unless otherwise specified herein, all chemicals shall be Reagent Grade or ACS Grade in accordance with MIL-STD-1218.

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4.6.1 Determination of water. Working in an area having a relative humidity of ≤3 percent (glovebox or dry room) fill a dried 10mm cylindrical spectrophotometric cell having near-infrared (NIR) silica windows with the manufactured electrolyte solution. This is best accomplished employing a glass pipette fitted with a rubber bulb. Stopper the cell with a ground glass or Teflon stopper. Place the filled cell in the sample beam of a Beckman Model 4240 or equivalent infrared spectrophotometer and an identical unfilled cell in the reference beam. Scan from 4000 to 2500 cm^{-1} at a rate of 600 cm^{-1}/min. Determine the percent transmission values (Po) and (P) for the peaks at approximately 3379 cm^{-1} and 2780 cm^{-1} as shown in Figure 1. Calculate the value of absorbance for each peak employing the following equation.

$$A = \log_{10} \frac{Po}{P}$$

If it is found that the absorbance of the 3379 cm^{-1} peak is greater than approximately 1.2, reanalyze the solution using a 5mm cell. If the absorbance is still greater than 1.2 repeat once again using a 2mm cell. Calculate the water concentration corresponding to the 2780 cm^{-1} peak (as measured in the 10mm cell) as follows:

$$\text{ppm } H_2O = \frac{\text{A at 2780 cm-1}}{0.009033}$$

Calculate the water concentration corresponding to the 3379 cm^{-1} peak as follows (using A = absorbance at 3379 cm^{-1}):

1) 10mm cell

$$\text{ppm } H_2O = \frac{A - 0.009}{0.006332}$$

2) 5mm cell

$$\text{ppm } H_2O = \frac{A - 0.005}{0.003169}$$

3) 2mm cell

$$\text{ppm } H_2O = \frac{A - 0.005}{0.000791}$$

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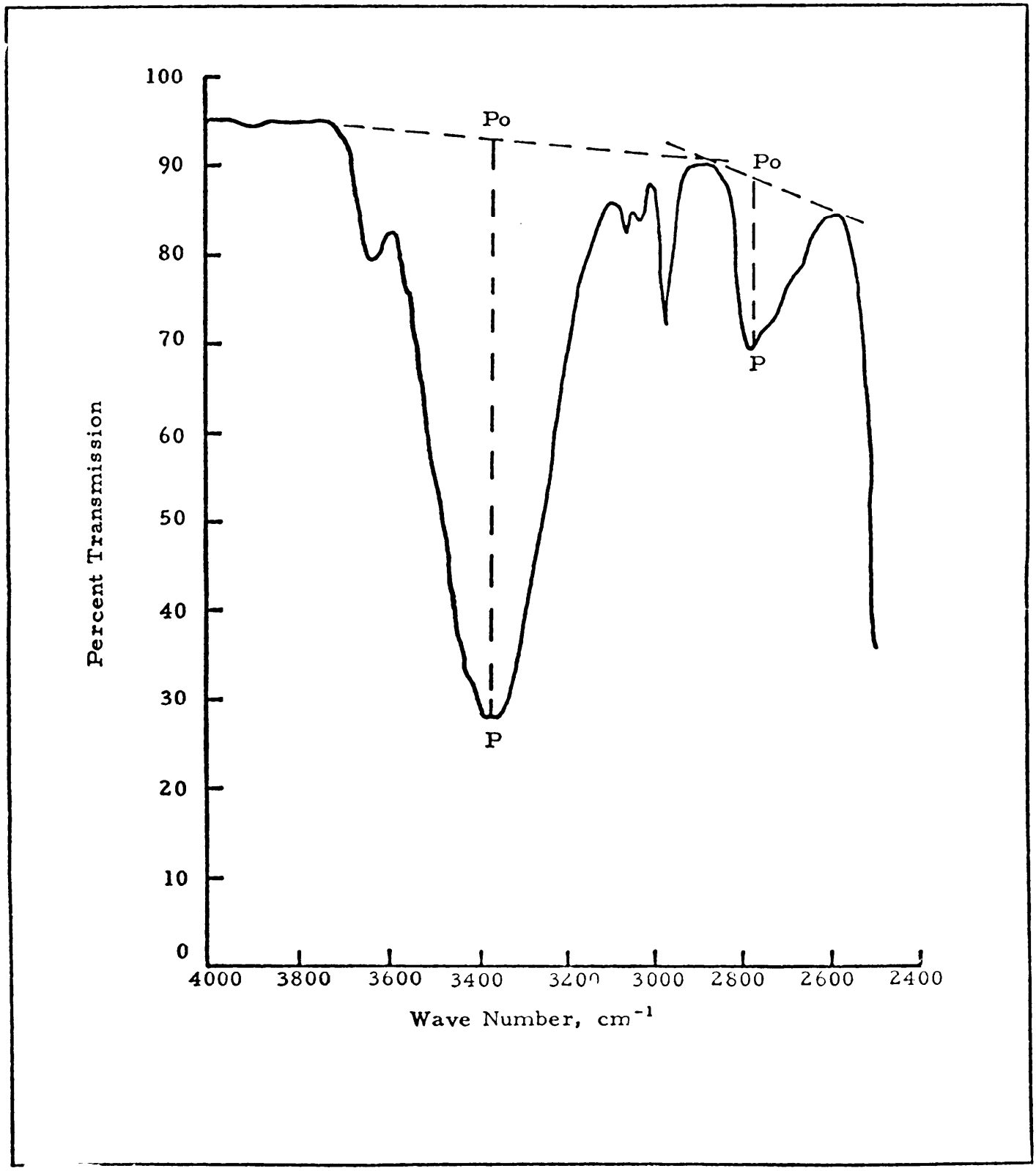


Figure 1. Typical Infrared Spectrum of a 1.0M $LiAlCl_4/SOCl_2$ Solution Showing How the Percent Transmission Values used in the Absorbance Calculations are Obtained

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subjected to the same [illegible] chemical or physical process

The total water concentration of the solution is obtained by summing the results:

ppm H_2O (Total) = ppm H_2O (2780 cm^{-1} peak)
+ ppm H_2O (3370 cm^{-1} peak)

4.6.2 Determination of iron. Working in an area having a relative humidity of $\leq$ 3 percent (glovebox or dry room), pipette 5.0 $\pm$ 0.1 ml of 1.0M $LiAlCl_4/SOCl_2$ solution into a dried 100 ml volumetric flask. If the work is being done in a glovebox, the flask may be stoppered and removed at this time. Slowly add 15-20 ml of deionized water and allow solution to stand for several minutes in order to dissipate the heat generated. Add 15-20 ml of deionized water to a similar volumetric flask and then add the following reagents, in the order listed, to each flask.

a. 30$\pm$0.5 mL of a 50-50 (by volume) mixture of methyl ethyl ketone and methanol.

b. 10$\pm$0.2 mL of 1:1 Hydrochloric Acid solution.

c. 5$\pm$0.1 mL of a 30 percent by weight Ammonium Thiocyanate (NH_4SCN) solution.

d. 5$\pm$0.1 mL of a 10 percent by weight Ammonium Thiosulfate $(NF_4)_2S_2O_3$) solution.

e. Dilute to mark with deionized water and mix well.

Fill a 0.75 inch diameter spectrophotometric test tube (Fisher Scientific Catalog No. 7-144-40 or equivalent) with each of the two solutions.

Read the absorbance of the test solution at 475 nm versus the blank employing a Bausch & Lomb Spectronic 20 or equivalent spectrophotometer.

Calculate the concentration of iron as follows:

$$X = \frac{Abs - 0.004}{0.003243}$$

Where

X = Iron concentration, mg
Abs = absorbance at 475 nm

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4.6.3 Determination of density. Working in an area having a relative humidity of ≤3 percent (glovebox or dry room) fill a dried, tared 25 ml volumetric flask to the volume mark with the manufactured electrolyte solution. Stopper the flask and weigh to 0.1 gram. Measure the temperature of the electrolyte solution to 0.5°C with a glass thermometer. Calculate the density of the electrolyte solution as follows:

$$d = \frac{G - T}{25}$$

Where,

d = solution density, g/mL
G = gross weight of volumetric flask, gm
T = Tare weight of volumetric flask, gm

Correct the density value to 24°C using the following equation:

$$d_{24} = 0.001631\ (T-24) + dT$$

Where

d_{24} = solution density at 24°C, g/mL
T = measured temperature of solution, °C
dT = measured solution density at temperature (T), g/mL

5. PACKAGING

5.1 Packing. (see 6.4)

5.1.1 Level C. Unless otherwise specified the electrolyte solution shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use.

5.2 Marking. Unless otherwise specified, the shipping containers shall be marked in accordance with MIL-STD-129, and shall include the lot number.

6. NOTES

6.1 Ordering data. See MIL-A-48078

6.2 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description

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of the proposed method should be submitted thru the Contracting Officer to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, Dover, NJ 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.3 Molar. A concentration in which one molecular weight in grams (one mole) of a substance is dissolved in one liter of solution. The Thionyl Chloride Electrolyte solution contains a five mole percent excess of Lithium Chloride which will cause a small amount of insoluble material.

6.4 Preparation and storage. The Thionyl Chloride Electrolyte solution is prepared and stored in a working area having a relative humidity of $\leq$5 percent (glovebox or dry room). The solution is prepared in a glass vessal or glass lined vessal, and stored in a dry room. Record the following:

Electrolyte Batch Number
Lithium Chloride manufacturer and lot number.
Aluminum Chloride manufacturer and lot number.
Thionyl Chloride manufacturer and lot number.
Date of Electrolyte manufactured.

Custodian:
Army-AR

Preparding Activity:
Army-AR

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-63420 (AR) THIONYL CHLORIDE ELECTROLYTE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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